

Fleet-Wide Condition Monitoring via Shared Nominal Representations and Local Radius Calibration

Enabling OEM-Centralized, Fleet-Deployed CBM

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Abstract

Objectives: Transportation operators deploying condition monitoring across large fleets of heterogeneous assets rarely have fault labels, time for per-asset representation learning, or long field calibration campaigns. This paper addresses how a single nominal representation can be deployed fleet-wide on day one, without retraining a model or tuning a threshold for every asset.

Methods: A canonical nominal representation, a Conv1D autoencoder, is trained once offline on simulated healthy signals and transferred unchanged to every asset. Each asset then estimates its operational radius, an empirical residual quantile, from a short stream of unlabeled local observations, without retraining the shared model. The method is evaluated on four Case Western Reserve University (CWRU) bearing conditions, spanning healthy, ball-fault, inner-race, and outer-race windows, against an oracle calibrated on each condition’s complete healthy dataset. Three calibration strategies are compared: direct threshold transfer, a complete-data oracle radius, and the proposed online radius estimated from only the first 100 healthy windows. The deployment principle is further validated on four independently monitored bearings from the NASA IMS run-to-failure dataset and compared with autoencoders trained separately on each bearing’s early-life data.

Findings: Direct transfer of the absolute simulator threshold produced complete false alarms in the field, confirming that the representation, not its absolute score scale, transfers. Using only a 100-window online calibration, the shared representation reached an average healthy false-alarm rate of 3.09%, ball-fault detection of 99.69%, and 100% detection of inner- and outer-race faults, with the radius converging after 51.0 ± 12.6 windows, approaching the asset-specific reference. On NASA IMS, the shared representation detected persistent departure in all four bearings, on average 83.25 hours before the documented end of the experiment, compared with 47.89 hours for the per-asset early-life autoencoders, while eliminating per-asset representation training.

Novelty: The contribution is a fleet-deployment methodology, not a new anomaly-detection architecture: it separates representation learning, performed once and shared fleet-wide, from decision-boundary calibration, performed independently by each asset.

Practical Applications: A shared representation can be developed and maintained centrally, for example by an original equipment manufacturer, and distributed unchanged across an operator’s fleet, while each installed asset requires only seconds of unsupervised local calibration and lightweight edge inference, removing the need for fault labels or per-asset model training during rollout.

Keywords: fleet-wide condition monitoring, shared nominal representation, simulation-to-field model transfer, local threshold calibration, run-to-failure monitoring, condition-based maintenance

1 Introduction

Modern transportation systems increasingly rely on condition-monitoring technologies to supervise large populations of operational assets, including rolling-element bearings, axleboxes,

traction motors, gearboxes, pumps, actuators, vehicles, and spatially indexed infrastructure segments Jardine et al. 2006; Zhao et al. 2019. Although machine-learning research has produced increasingly capable anomaly detectors Chandola et al. 2009; Zhao et al. 2019, the deployment problem remains largely unresolved: a detector that performs well for one asset under one calibrated operating condition does not automatically scale to hundreds or thousands of independently installed units.

At fleet rollout, several constraints arise simultaneously. First, representative fault examples are rarely available. Second, waiting months or years for failures to accumulate is operationally unacceptable. Third, installation-specific factors such as sensor coupling, operating point, mounting, and environmental noise alter the numerical scale of anomaly scores between assets Randall and Antoni 2011. Finally, training, calibrating, validating, and maintaining a separate representation model for every physical unit creates an engineering burden that grows directly with fleet size.

The practical question is therefore not simply

Q1. *How can an anomalous condition be detected?*

but rather

Q2. *How can one nominal representation be deployed on day one across a heterogeneous fleet while retaining performance close to an ideal asset-specific solution, without per-asset representation learning or manual threshold tuning?*

This paper addresses Q2 by separating representation engineering from deployment calibration. A canonical nominal representation is learned once, offline, from simulated healthy observations and distributed unchanged across the fleet. Each deployed asset estimates only its own operational radius from a short stream of unlabeled local observations. The shared model is never retrained, fine-tuned, or adapted in the field.

From a broader perspective, this separation can be interpreted as a local inverse calibration problem. Representation learning is performed once to recover a canonical nominal representation, whereas deployment requires each asset only to identify its own operational radius around that fixed representation. This interpretation follows the inverse-problem and quotient-space formulations introduced in our previous work Di Santi 2026b; Di Santi 2026a, while the present paper focuses on their engineering realization for scalable fleet deployment.

The transferable object is therefore not an anomaly threshold but the canonical nominal representation itself. Absolute reconstruction-error thresholds learned in simulation generally do not transfer because their numerical scale depends on the simulator, preprocessing, and residual distribution. Instead, the proposed methodology transfers the shared representation and a fleet-wide nominal coverage, while each asset estimates locally the empirical residual quantile that realizes that coverage.

This formulation also changes the relevant experimental comparison. The ideal reference is not another globally shared detector but an operationally unrealistic situation in which every asset has its complete healthy dataset available for independent representation learning and calibration. Evaluating Q2 requires answering a third question:

Q3. *How much detection performance is lost when many asset-specific models are replaced by one simulator-trained representation and a few seconds of local calibration?*

We answer Q3 first under controlled operating-condition heterogeneity using four CWRU conditions Smith and Randall 2015, and then evaluate its extension to independently monitored physical units undergoing naturally evolving degradation using the NASA IMS run-to-failure dataset.

The asset-specific reference methods are trained and calibrated using the complete healthy dataset of each operating condition. In contrast, the proposed method uses no field observations for representation learning and estimates the local operational radius from only the first 100 healthy windows, corresponding to approximately ten seconds of signal.

The experimental results show that the cost of scalable deployment is small. Asset-specific autoencoders reach 100% detection across the evaluated ball-, inner-race-, and outer-race-fault families. The unchanged simulator-trained shared model, combined with online local-radius estimation, reaches an average ball-fault detection rate of 99.69% and 100% detection of inner- and outer-race faults. The operational radius converges after 51.0 ± 12.6 windows on average. Direct transfer of the absolute simulator threshold, however, produces 100% healthy false alarms, confirming that the representation transfers whereas its absolute score scale does not.

The contribution of this paper is therefore a deployment methodology rather than a new anomaly- detection architecture. Although a Conv1D autoencoder is used as the experimental representation, the deployment principle is representation-agnostic. The principal contributions are:

1. a fleet-wide architecture that separates one shared canonical nominal representation from independently calibrated local operational radii;
2. a direct simulation-to-field deployment in which the representation is transferred unchanged without field-data retraining;
3. an online empirical-quantile calibration procedure that estimates an operational radius from a short unlabeled warm-up;
4. a convergence criterion separating finite cold-start calibration from subsequent degradation monitoring;
5. an evaluation against ideal asset-specific detectors trained and calibrated with complete healthy datasets; and
6. an edge-oriented deployment model in which the representation may be engineered and maintained centrally while deployed assets perform only lightweight local calibration.

The methodology is independent of asset type and sensing modality. It requires only a nominal representation producing a scalar anomaly score, repeated observations from each operational unit, and a local procedure for estimating the acceptance boundary. Rolling-element bearings are used here as the first controlled validation domain rather than as a limitation of the proposed deployment principle.

The remainder of the paper is organized as follows. Section 2 reviews condition monitoring, one-class anomaly detection, adaptive calibration, and fleet deployment. Section 3 introduces the deployment methodology. Section 4 presents the experimental protocol. Section 5 reports the deployment results. Section 7 discusses industrial implications and limitations. Section 6 summarizes the mathematical interpretation.

2 Related Work

Condition-based maintenance (CBM) has become a central component of modern transportation asset management, with condition monitoring providing the operational evidence required to supervise large fleets of heterogeneous assets Jardine et al. 2006; Chandola et al. 2009; Zhao et al. 2019. Although sensing modalities differ across applications, the computational objective remains the same: to characterize nominal operation and detect persistent departures that may indicate degradation.

Most machine-learning approaches formulate this problem as supervised fault classification, requiring representative fault examples during training. While effective on benchmark datasets, these methods assume that deployment conditions remain close to those observed during model development.

One-class methods, including one-class SVM Schölkopf et al. 1999, neural-network novelty detection Bishop 1994, and autoencoder-based anomaly detection Hinton and Salakhutdinov 2006; Sakurada and Yairi 2014, reduce the dependence on fault labels by learning only nominal behaviour. More recent representation-learning approaches further improve anomaly detection by constructing latent spaces in which healthy operation occupies a compact region while degraded behaviour becomes more easily separable.

However, most existing work focuses on improving anomaly detectors rather than addressing the deployment problem itself: how a single nominal representation can be initialized, calibrated, and maintained across heterogeneous fleets.

Adaptive thresholds have been widely used to compensate for changing operating conditions. However, unrestricted adaptation risks absorbing progressive degradation into the nominal baseline. This motivates separating calibration from monitoring. In the proposed methodology, adaptation is limited to a finite cold-start phase during which each asset estimates its local operational radius. Once convergence is achieved, the shared representation remains fixed and subsequent departures become diagnostic evidence rather than input for further adaptation.

The proposed deployment strategy is also related to invariant representation learning Bronstein et al. 2021 and anomaly detection under distribution shift **li2023invariant**, both of which seek representations that are insensitive to nuisance variability. Our contribution, however, lies at a different level. Rather than proposing a new representation-learning architecture, we address the deployment problem by combining a shared nominal representation with local operational-radius calibration, eliminating per-asset representation learning while preserving a common nominal coverage across the fleet.

This viewpoint is also consistent with the inverse-problem and quotient-space formulations introduced in our previous work Di Santi 2026b; Di Santi 2026a, although the present paper focuses exclusively on their engineering realization for scalable fleet deployment.

The methodology is independent of sensing modality and asset type. The CWRU bearing benchmark Smith and Randall 2015 is used as a controlled validation domain rather than as a limitation of the proposed deployment principle.

3 Method

This section formalizes the proposed deployment methodology, which separates a fleet-wide shared nominal representation from asset-specific operational-radius calibration.

3.1 Deployment architecture: shared representation, local radius

Let $x_{k,t} \in \mathcal{X}$ denote an observation window acquired from operational unit k at time t . In the bearing experiments, $x_{k,t}$ is instantiated as a fixed-length vibration window. A shared offline-trained representation produces an anomaly score

$$r_{k,t} = r(x_{k,t}). \quad (1)$$

The representation is shared across the fleet, whereas the decision boundary is estimated independently for each asset. Operational unit k maintains a local operational radius $\tau_k(t)$ and raises an alarm whenever

$$r_{k,t} > \tau_k(t). \quad (2)$$

The deployment methodology therefore follows three principles:

1. one shared representation for the fleet;
2. local calibration for each asset; and
3. no per-asset representation retraining.

3.2 Representation instantiation

The proposed deployment methodology is instantiated with a Conv1D autoencoder $A_\phi = D_\phi \circ E_\phi$ trained exclusively on nominal signals. The encoder maps a 1200-sample window through three Conv1D layers (32, 64, and 128 filters) to a 16-dimensional latent code with an ℓ_1 activity penalty, while the decoder mirrors this architecture using Conv1DTranspose layers. The network contains 847,601 trainable parameters and is optimized with Adam, mean-squared reconstruction loss, early stopping, and learning-rate decay.

The anomaly score is the reconstruction residual

$$r(x) = \|x - A_\phi(x)\|_2^2. \quad (3)$$

The contribution of this paper lies in the deployment methodology rather than in a new representation-learning architecture. The Conv1D autoencoder serves as the nominal representation throughout the experimental validation, while the deployment principles introduced here are independent of this particular training procedure.

3.3 Shared nominal coverage and local empirical radius

The fleet-wide calibration parameter is a target nominal coverage level $1 - \alpha$, rather than an absolute anomaly-score threshold. In the present experiments, $1 - \alpha = 0.98$.

For operational asset k , let $\hat{Q}_{k,t}(1 - \alpha)$ denote the empirical $(1 - \alpha)$ -quantile of the local residuals observed during initialization up to time t . The local operational radius is

$$\tau_k(t) = \hat{Q}_{k,t}(1 - \alpha). \quad (4)$$

The simulator threshold

$$\tau_{MC} = \hat{Q}_{MC}(1 - \alpha) \quad (5)$$

is retained only for comparison. Its absolute numerical value is not used by the proposed field-deployment procedure. What is shared across the fleet is the target nominal coverage, while each asset identifies the residual value that realizes that coverage locally.

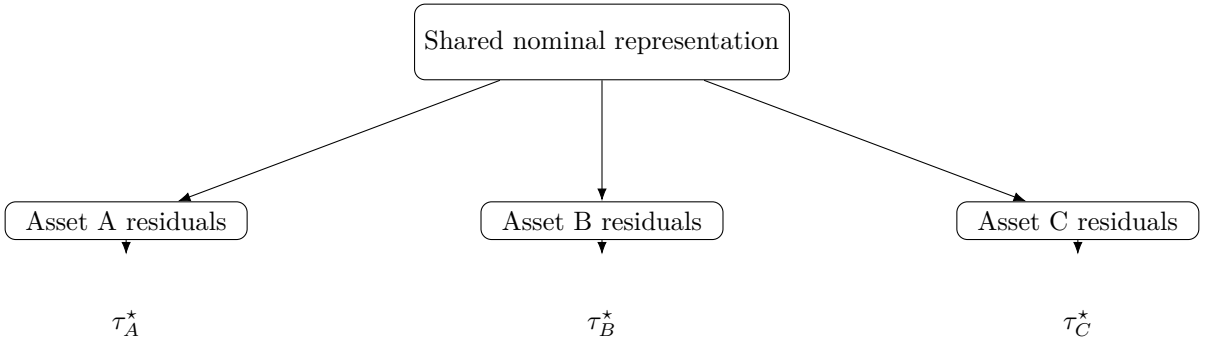


Figure 1: The same shared nominal representation produces reconstruction residuals for every deployed asset. Only the local empirical residual distribution changes across installations. Each asset independently estimates its own operational radius τ_k^* , while the representation remains unchanged.

3.4 Operational-radius convergence

During cold start, each asset updates Eq. 4 using the incoming nominal residuals. Calibration is declared when the operational radius stabilizes.

For a look-back interval L , define the convergence statistic

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t - L)|}{\max(\tau_k(t - L), \epsilon)}, \quad (6)$$

where $\epsilon > 0$ prevents division by zero.

Calibration is declared when

$$C_k(t) < \varepsilon_C \quad (7)$$

for m consecutive updates. The converged operational radius is then

$$\tau_k^* = \tau_k(t_{\text{conv}}). \quad (8)$$

The key hypothesis evaluated in this paper is: *Under stable nominal operation, every asset converges rapidly to its own operational radius around a fixed shared representation.*

After convergence, τ_k^* is frozen (or only robustly corrected) to prevent progressive degradation from being absorbed into the nominal baseline. The convergence criterion therefore marks the transition from calibration to operational monitoring.

3.5 Guarded updates

For deployments requiring continued adaptation beyond the controlled warm-up, a guarded update rule can be applied to the local baseline. Let $\delta > 0$ define a guard band. Local statistics are updated only when

$$r_{k,t} < \tau_k(t) - \delta. \quad (9)$$

The three operating regions are

$$r_{k,t} < \tau_k(t) - \delta \quad \Rightarrow \quad \text{safe update}, \quad (10)$$

$$\tau_k(t) - \delta \leq r_{k,t} \leq \tau_k(t) \quad \Rightarrow \quad \text{observe, do not update}, \quad (11)$$

$$r_{k,t} > \tau_k(t) \quad \Rightarrow \quad \text{anomaly}. \quad (12)$$

This rule requires only a residual buffer (or running robust statistics), the current operational radius, and the convergence state.

3.6 Deployment states

Each asset progresses through three deployment states:

1. **Cold start:** local estimation of the operational radius;
2. **Converged monitoring:** τ_k^* is fixed and anomaly detection is active;
3. **Operational departure:** persistent exceedances of the converged operational radius.

In practice, alarms may require q exceedances within the last p observations to avoid reacting to isolated impulses.

3.7 Scope of the Method

The proposed methodology is independent of asset type and sensing modality. Throughout the paper, the index k denotes an independently monitored operational unit, such as a bearing, traction motor, gearbox, pump, actuator, vehicle, or infrastructure segment.

The methodology requires only (i) a shared nominal representation, (ii) repeated observations from each operational unit, and (iii) local estimation of the operational radius.

The CWRU bearing benchmark is used solely as a controlled validation domain.

4 Experimental Design

The experimental protocol evaluates the proposed deployment methodology rather than the underlying representation-learning architecture. Across the five experimental stages, we measure:

- false-alarm rate during cold start;
- number of accepted samples to operational-radius convergence;
- variability of τ_k^* across assets and regimes;
- post-convergence detection rate;
- sensitivity to buffer size and convergence parameters;
- performance of absolute-threshold transfer versus shared-coverage calibration;
- cost of per-asset retraining avoided by the shared-model design.

Stage 1 — Controlled multi-asset simulation. Four bearing streams are generated with a physics-motivated nominal simulator using 1200-sample windows, a 30 Hz shaft frequency, and bounded variation in amplitude, phase, speed, offset, and noise. The shared model is trained on 4,000 simulated nominal windows, with 1,000 additional nominal windows reserved for validation. Faults are introduced only during evaluation. A separate synthetic test set contains 750 windows per regime. Each simulated asset begins in nominal operation and independently calibrates its radius. Asset A remains nominal; Assets B–D transition after convergence to outer-race, inner-race, and ball-fault regimes.

This stage measures: (i) samples to convergence; (ii) false alarms before and after convergence; (iii) the separation between $r_{k,t}$ and τ_k^* after fault onset; and (iv) whether each nominal asset reaches a stable but asset-specific radius.

Stage 2 — Direct simulation-to-field deployment on CWRU. The nominal representation trained in Stage 1 is transferred unchanged to the Case Western Reserve University (CWRU) Drive-End dataset. No CWRU observation is used for representation learning, retraining, or fine tuning.

Each operating condition is treated as an independent deployment environment. The first 100 healthy windows are used only to estimate the local 98th-percentile operational radius, after which fault observations are evaluated using the unchanged shared model.

This stage evaluates whether (i) the learned representation transfers directly from simulation to field, and (ii) local empirical calibration compensates for the non-transferability of the absolute simulator threshold.

The four CWRU operating conditions provide a controlled evaluation of score-scale heterogeneity and local calibration rather than four physically distinct fleet assets. Validation on independent run-to-failure bearings is addressed separately in Stage 4 using the NASA IMS dataset.

Stage 3 — Reference comparison and calibration ablation. To quantify the cost of scalability, we construct an intentionally favorable but operationally non-scalable reference condition. For each CWRU operating condition, three detectors are trained and calibrated independently using the complete available healthy dataset: (i) a Conv1D autoencoder; (ii) a one-class SVM operating on PCA-reduced standardized windows; and (iii) an Elliptic Envelope operating on PCA-reduced standardized windows.

Each detector uses the empirical 98th percentile of its own healthy anomaly-score distribution as the decision boundary. Fault observations are never used for model fitting or threshold selection. These experiments represent an operationally unrealistic upper bound in which complete nominal data and asset-specific model development are available for every monitored unit.

Using the unchanged simulator-trained autoencoder, we compare three deployment strategies:

1. **Direct threshold transfer:** apply the absolute synthetic p98 residual threshold unchanged to each CWRU condition;
2. **Field oracle:** estimate the p98 operational radius using the complete healthy field dataset of each condition, without retraining the shared representation; and
3. **Online local convergence:** estimate the p98 operational radius progressively from only the first 100 unlabeled healthy windows.

This ablation separates representation transfer from decision-boundary transfer. The representation is identical in all three conditions; only the information available for local calibration changes.

Stage 4 — Multi-unit run-to-failure validation on NASA IMS. A second nominal representation is constructed for the IMS bearing family using simulated healthy observations matched to the dataset sampling and operating characteristics. This model is trained once and then transferred unchanged to the independently monitored bearings within the IMS experiments. Each bearing estimates its own operational radius from an early-life unlabeled segment and subsequently enters post-convergence monitoring.

This stage evaluates whether the proposed deployment principle extends from heterogeneous CWRU operating conditions to physically distinct bearing units and naturally evolving degradation trajectories. We report: (i) samples to operational-radius convergence; (ii) variability of the converged radius across bearings; (iii) healthy-period false-alarm rate; (iv) first persistent departure from the frozen radius; and (v) alarm lead time relative to the documented end of the experiment.

Stage 5 — Edge cost. We report model size, parameter count, inference latency per 1200-sample window on CPU, local state memory, and the reduction in transmitted data when only scores and selected anomalous windows are sent rather than the continuous raw signal.

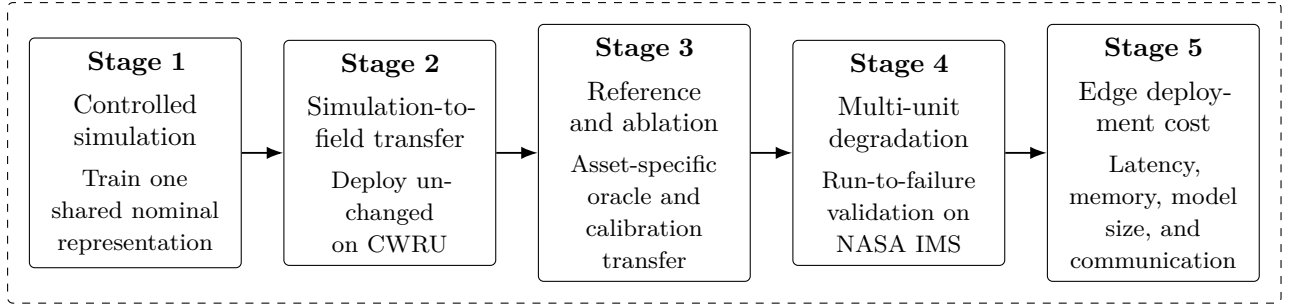


Figure 2: Deployment-oriented experimental protocol. The experiments progress from controlled representation learning to direct field transfer, ideal-reference comparison, multi-unit run-to-failure validation, and measurement of the edge-deployment footprint. Across deployment stages, the nominal representation remains fixed while each monitored unit estimates its own local operational radius.

5 Results

The experiments address three complementary deployment questions. First, the CWRU experiments establish the performance attainable under an ideal but operationally non-scalable condition in which separate detectors are trained and calibrated for each operating condition. Second, they evaluate whether a simulator-trained shared representation, combined only with short local calibration, can approach that ideal performance without field-model retraining. Third, the NASA IMS run-to-failure experiment evaluates whether the same deployment principle extends to independently monitored physical units undergoing naturally evolving degradation.

The CWRU evaluation includes four operating conditions corresponding to 1730, 1750, 1772, and 1797 RPM. Across these conditions, the experiments contain 1,414 healthy windows, 1,621 ball-fault windows, 1,621 inner-race-fault windows, and 2,846 outer-race-fault windows. All windows contain 1,200 samples and are processed identically across detectors. The NASA IMS evaluation subsequently considers four independently monitored bearings from Experiment 2, each represented by 984 chronologically ordered recordings, of which the first 100 are reserved for early-life calibration.

5.1 Ideal asset-specific reference

Table 1 reports the performance of three detectors trained independently for each CWRU operating condition using the complete healthy dataset available for that condition. These experiments constitute an oracle reference rather than a deployable fleet architecture, assuming complete nominal data and independent training and calibration for every operating condition.

Since each detector is calibrated at the empirical 98th percentile of its healthy anomaly-score distribution, the healthy false-alarm rate is approximately 2% by construction. The comparison therefore evaluates the diagnostic quality of the anomaly score rather than threshold estimation.

Table 1: Ideal asset-specific detector performance on CWRU. Values are mean $\pm$ standard deviation across the four operating conditions. Each detector is trained and calibrated separately using the complete healthy dataset of the corresponding condition.

Detector	Healthy FAR (%)	Ball DR (%)	Inner-race DR (%)	Outer-race DR (%)
Asset-specific AE	2.29 $\pm$ 0.12	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
Asset-specific OC-SVM	2.29 $\pm$ 0.12	3.09 $\pm$ 6.17	25.05 $\pm$ 0.13	2.71 $\pm$ 3.15
Asset-specific Elliptic Envelope	2.29 $\pm$ 0.12	0.62 $\pm$ 1.23	14.43 $\pm$ 10.63	1.76 $\pm$ 3.15

The asset-specific autoencoder achieves 100% detection for all evaluated fault families under all four operating conditions. In contrast, the classical one-class detectors reproduce the target false-alarm rate but provide little diagnostic separation, with average ball-fault detection rates of 3.09% (OC-SVM) and 0.62% (Elliptic Envelope).

These results indicate that nominal calibration alone is insufficient. Although all detectors achieve the prescribed healthy false-alarm rate, only the reconstruction-based representation consistently separates healthy and faulty operation. The asset-specific autoencoder is therefore used as the oracle reference for evaluating the cost of shared-model deployment.

5.2 Repeated-run stability of the asset-specific autoencoder

To assess sensitivity to stochastic optimization, the complete training and evaluation procedure was repeated five times using different random initializations. The architecture, preprocessing, datasets, training protocol, and calibration procedure remained unchanged.

Table 2 shows identical healthy false-alarm and fault detection performance across all operating conditions. The only quantity showing measurable variability is the absolute reconstruction threshold estimated from the healthy residual distribution, whose numerical value changes slightly between training runs without affecting diagnostic performance.

Table 2: Repeated-run stability of the asset-specific autoencoder over five independent random initializations. Detection performance is unchanged across runs, whereas the absolute reconstruction threshold varies slightly.

RPM	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Threshold
1730	2.23 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	0.649 ± 0.021
1750	2.23 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	0.759 ± 0.227
1772	2.23 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	0.725 ± 0.148
1797	2.46 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	0.459 ± 0.026

5.3 Direct simulation-to-field deployment

The shared autoencoder is trained once using simulated healthy observations and is transferred unchanged to all four CWRU conditions. No CWRU observation is used to retrain, fine-tune, or otherwise modify the model or its preprocessing transformation.

Three calibration conditions are evaluated:

1. **Direct threshold transfer**, in which the absolute p98 residual threshold estimated in simulation is applied unchanged in the field;
2. **Field oracle**, in which the p98 operational radius is computed using the complete healthy field dataset for each condition; and
3. **Proposed online convergence**, in which the local p98 operational radius is estimated progressively from only the first 100 unlabeled healthy windows.

Table 4 reports the complete results.

Table 3: Simulation-to-field deployment of the shared autoencoder on CWRU. FAR denotes the healthy false-alarm rate and DR the fault detection rate. The field-oracle radius uses every available healthy window; the online radius uses only the first 100 healthy windows. A dash indicates that convergence is not applicable because the threshold is computed directly rather than estimated sequentially.

RPM	Calibration	τ	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Healthy windows	Warm-up (windows)	Convergence (windows)
1730	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	404	0	–
1730	Field oracle	0.2005	2.23	100.00	100.00	100.00	404	404	–
1730	Proposed online convergence	0.1984	2.97	100.00	100.00	100.00	404	100	38
1750	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	404	0	–
1750	Field oracle	0.1786	2.23	100.00	100.00	100.00	404	404	–
1750	Proposed online convergence	0.1759	2.97	100.00	100.00	100.00	404	100	58
1772	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	403	0	–
1772	Field oracle	0.1997	2.23	100.00	100.00	100.00	403	403	–
1772	Proposed online convergence	0.2072	0.50	100.00	100.00	100.00	403	100	43
1797	Direct threshold transfer	0.0979	100.00	100.00	100.00	100.00	203	0	–
1797	Field oracle	0.2474	2.46	96.80	100.00	100.00	203	203	–
1797	Proposed online convergence	0.2364	5.91	98.77	100.00	100.00	203	100	65

Direct transfer of the absolute simulator threshold fails under every operating condition. Its 100% apparent fault-detection rate is meaningless because it simultaneously classifies 100% of the healthy field windows as anomalous. The shared representation therefore transfers, but its absolute reconstruction-error scale does not.

The field-oracle calibration restores the target false-alarm level, obtaining an average healthy FAR of $2.29 \pm 0.12\%$. With this complete knowledge of each field condition, the unchanged simulator-trained model obtains $99.20 \pm 1.60\%$ ball-fault detection and 100% detection of both inner- and outer-race faults.

The proposed online procedure reaches comparable performance using only the first 100 healthy windows. Across the four operating conditions, it achieves an average healthy FAR of $3.09 \pm 2.22\%$, ball-fault detection of $99.69 \pm 0.62\%$, and 100% detection of both inner- and outer-race faults. The online operational radius converges after 51.0 ± 12.6 windows, with observed convergence indices ranging from 38 to 65 windows.

Table 4: Simulation-to-field deployment of the shared autoencoder on CWRU. FAR denotes the healthy false-alarm rate and DR the fault detection rate. The field-oracle radius uses every available healthy window; the online radius uses only the first 100 healthy windows. A dash indicates that convergence is not applicable because the threshold is computed directly rather than estimated sequentially.

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5.4 Cost of scalable deployment

Table 5 compares the proposed shared deployment with an ideal but operationally non-scalable asset-specific reference. The comparison intentionally favors the baseline: the oracle detector is trained and calibrated using the complete healthy dataset of each operating condition, whereas the proposed method estimates its local operational radius from only the first 100 healthy windows and performs no field-model retraining.

Table 5: Ideal but non-scalable asset-specific deployment versus the proposed fleet-scalable deployment. Values are mean $\pm$ standard deviation across the four CWRU operating conditions.

Deployment strategy	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Field model training	Healthy field data used
Asset-specific AE oracle	2.29 ± 0.12	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	One model per condition	Complete dataset
Shared AE with field-oracle radius	2.29 ± 0.12	99.20 ± 1.60	100.00 ± 0.00	100.00 ± 0.00	None	Complete dataset
Proposed shared AE with online radius	3.09 ± 2.22	99.69 ± 0.62	100.00 ± 0.00	100.00 ± 0.00	None	100-window warm-up

Replacing four independently trained autoencoders with one simulator-trained shared model produces no observed loss for inner- or outer-race detection and less than one percentage point of average difference for ball-fault detection. The principal cost appears in healthy false alarms, whose mean increases from 2.29% to 3.09%. Most of this increase is concentrated in the 1797 RPM condition (5.91%), whereas the remaining conditions range between 0.50% and 2.97%.

These results support the central deployment claim of this paper. A single simulator-trained canonical representation, combined with short local operational-radius calibration, achieves near-oracle detection performance without asset-specific representation learning, substantially reducing deployment complexity.

5.5 Operational-radius convergence

Figure 3 shows the sequential estimation of the local operational radius under the fixed shared representation. For each CWRU operating condition,

$$\tau_k(t) = \mu_k(t) + z_{MC}\sigma_k(t), \quad (13)$$

where z_{MC} is transferred from the simulated nominal residual distribution. The representation and standardized coverage remain fixed; only the local residual statistics are estimated from incoming healthy observations.

The first 100 windows constitute the prescribed cold-start calibration interval. Across all four operating conditions, the convergence criterion is satisfied between windows 38 and 65, before the calibration budget is exhausted. The operational radii at window 100 are 0.2393, 0.2188, 0.2540, and 0.2777 for 1730, 1750, 1772, and 1797 RPM, respectively.

Cold-start convergence across CWRU operating conditions

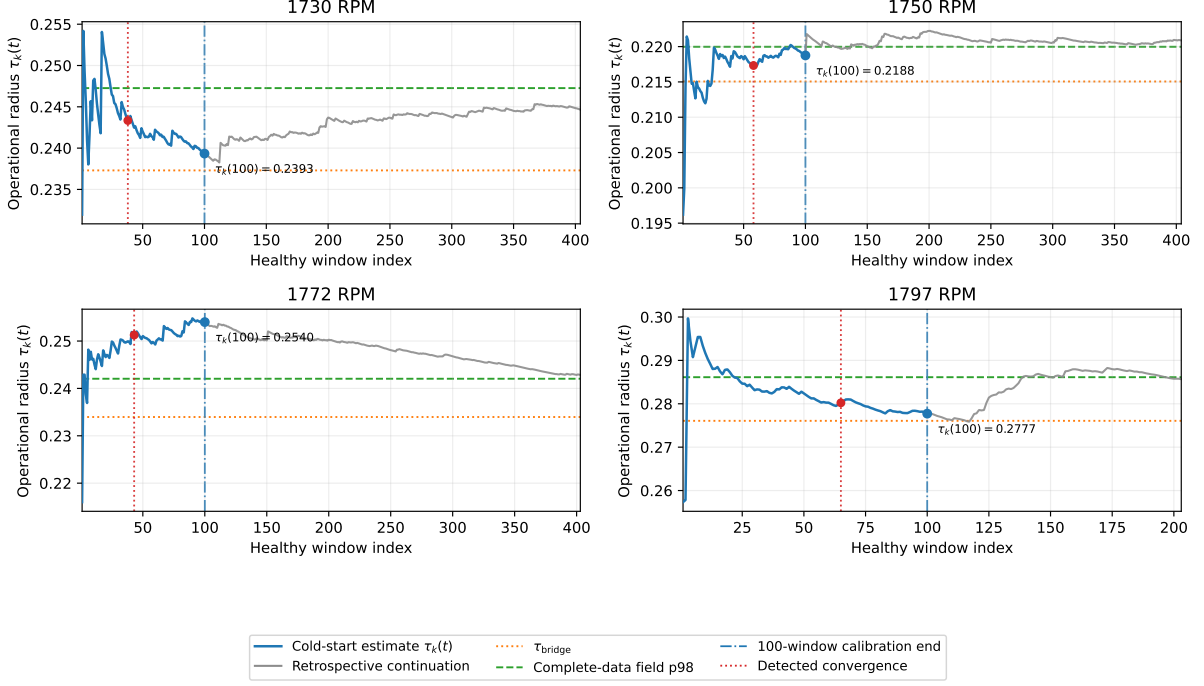


Figure 3: Cold-start convergence and retrospective evolution of the local operational radius across the four CWRU operating conditions. The blue segment shows the sequential radius estimate during the 100-window calibration interval used by the deployed detector. The red dotted line marks the detected convergence index, and the blue dash-dotted line marks the end of the prescribed cold-start interval. The gray segment is a retrospective cumulative estimate obtained by continuing to incorporate the remaining known-healthy observations; it is shown only to characterize the subsequent evolution of the estimator and is not used to update the deployed decision boundary. Horizontal lines indicate the transferred standardized radius τ_{bridge} and the complete-data field p98 reference.

The gray trajectories shown after the calibration interval are retrospective only. They illustrate how the estimated operational radius would evolve if additional observations, known to be healthy in the benchmark, continued updating the local statistics. They are not used by the deployed detector, whose operational radius is frozen after convergence.

The observed post-calibration fluctuations are expected because the healthy residual distribution is not perfectly stationary. Later nominal observations modify the estimated residual mean, variance, or both, causing the cumulative estimate to increase or decrease without implying degradation. This behaviour motivates separating finite calibration from subsequent monitoring.

Operationally, the result demonstrates that an asset-specific acceptance boundary can be established before the complete healthy distribution is available. The complete-data p98 reference assumes prior access to all healthy observations, whereas the proposed procedure reaches a usable local radius during a short initialization interval in every evaluated operating condition.

The present experiment assumes that the initialization data are representative of nominal operation. Detecting assets that are already degraded at installation, or determining that no stable nominal radius can be established, remains outside the present scope and constitutes future work.

5.6 Robustness to local initialization

A practical deployment question is whether local calibration depends critically on the particular healthy observations available during installation. To evaluate this sensitivity, the shared simulator-trained representation, preprocessing, and all model parameters were kept fixed. Only the initialization observations were varied.

For each CWRU operating condition, five independent warm-up sequences were generated by randomly sampling 100 healthy windows with different random seeds. Each sequence independently estimated the local operational radius, after which fault detection was evaluated using the unchanged shared representation.

Table 6: Robustness of online operational-radius estimation under different random warm-up samples. The shared simulator-trained representation remained fixed throughout the experiment. Only the 100 healthy initialization windows used for local calibration were randomly changed. Values are mean $\pm$ standard deviation over five random seeds.

RPM	τ_{online}	Healthy FAR (%)	Ball DR (%)	Inner DR (%)	Outer DR (%)	Convergence (windows)
1730	0.2905 ± 0.0041	4.08 ± 2.04	99.75 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	45.6 ± 13.2
1750	0.3080 ± 0.0063	3.49 ± 2.47	99.95 ± 0.11	100.00 ± 0.00	100.00 ± 0.00	54.2 ± 13.6
1772	0.3097 ± 0.0076	2.18 ± 1.66	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	49.4 ± 13.8
1797	0.3415 ± 0.0051	3.49 ± 2.44	96.16 ± 0.22	100.00 ± 0.00	100.00 ± 0.00	56.2 ± 15.0

Across all repetitions, the estimated operational radii remained close to their complete-data field-oracle values despite the different initialization samples. Detection performance was correspondingly stable: ball-fault detection remained between approximately 96% and 100%, while inner- and outer-race faults continued to be detected with essentially unchanged performance. The principal variability appeared in the healthy false-alarm rate, which depends on the particular empirical percentile estimated from only 100 initialization windows.

This behaviour is expected statistically because the empirical 98th-percentile is determined by only a few extreme observations. Small changes in those observations therefore produce modest changes in the estimated operational radius without materially affecting diagnostic separation.

These results show that the transferable object remains the shared canonical representation, whereas the local operational radius is consistently recovered from short unlabeled initialization data despite variations in the specific warm-up observations.

5.7 Multi-unit run-to-failure validation on NASA IMS

The CWRU experiments validate simulation-to-field transfer under controlled operating conditions. To evaluate whether the same deployment principle extends to independent physical units undergoing naturally evolving degradation, the shared nominal representation was further evaluated on the NASA IMS run-to-failure dataset.

Each NASA bearing is treated as an independent monitored asset. The simulator-trained autoencoder is transferred unchanged, and no NASA observations are used for representation learning, retraining, or parameter adaptation. Only a short early-life nominal segment is used to estimate the local operational radius.

Table 7 summarizes the recording-level results. The calibration exceedance rate is 2% by construction because the operational radius is defined as the empirical 98th percentile of the early-life scores. After calibration, a persistent departure is detected for all four bearings. The mean monitoring exceedance rate is 28.51%, and the first persistent departure occurs, on average, 83.25 hours before the documented end of the experiment.

Table 7: Recording-level run-to-failure validation on NASA IMS using the shared nominal representation and local operational-radius calibration. Lead time is measured relative to the documented end of the experiment.

Bearing	Calibration exceedance (%)	Persistent departure	Monitoring exceedance (%)	Lead time (h)
1	2.00	Yes	51.47	137.83
2	2.00	Yes	16.86	46.83
3	2.00	Yes	10.29	15.50
4	2.00	Yes	35.41	132.83
Mean	2.00	4/4	28.51	83.25

These results show that the proposed deployment methodology extends beyond controlled operating-condition transfer. A single shared nominal representation can be deployed across independently monitored physical units, requiring only local operational-radius calibration and no per-asset representation retraining.

5.7.1 Stability of local operational-radius estimation

A key deployment requirement is that the local calibration procedure should not depend critically on the particular nominal observations collected during installation. Therefore, the shared representation was kept fixed while multiple random initialization samples were used to estimate the local operational radius.

For each NASA bearing, five independent 100-window samples were drawn from the early-life calibration recordings. The shared representation and its preprocessing remained fixed; only the residual observations used to estimate the local operational radius were varied. Table 8 reports the resulting variability. The estimated radius remained stable across initialization samples, with low relative error and convergence within a limited number of calibration windows.

Table 8: Stability of local operational-radius estimation for the shared NASA representation under different warm-up initializations.

Bearing	τ mean	τ std	Relative error (%)	Convergence (windows)
1	0.0120	0.00013	0.92	48.4
2	0.0165	0.00010	2.07	39.0
3	0.0204	0.00050	2.08	62.2
4	0.0074	0.00009	1.82	45.6
Mean	0.0141	–	1.72	48.8

The stability experiment supports the central deployment hypothesis: the transferable object is the nominal representation, while each physical asset independently identifies its own local acceptance region. Variability between assets is therefore handled by calibration of the operational radius rather than by training separate representations.

5.7.2 Comparison with per-asset early-life training

To quantify the cost of eliminating per-asset representation learning, the proposed deployment was compared with an intentionally favorable reference. For each NASA bearing, an independent autoencoder and preprocessing transformation were trained using its first 100 recordings, repeating the training with five random initializations. The proposed method instead used one frozen simulator-trained representation for all bearings and estimated only a local operational radius.

Table 9 summarizes the recording-level comparison. Both approaches detect a persistent departure for every bearing and reproduce the prescribed 2% calibration exceedance. Monitoring exceedance rates are also similar. Under the present protocol, the shared representation detects the departure earlier for Bearings 1, 2, and 4, while both approaches produce nearly identical timing for Bearing 3.

Table 9: NASA IMS comparison between the proposed shared representation and autoencoders trained separately on the early-life data of each bearing. Asset-specific results are mean $\pm$ standard deviation over five training initializations. Shared-model values correspond to one fixed representation applied unchanged to every bearing. Lead time is measured relative to the documented end of the experiment.

Bearing	Deployment strategy	Departure detected (%)	Monitoring exceedance (%)	Lead time (h)
1	Asset-specific early-life AE	100.00 $\pm$ 0.00	53.08 $\pm$ 1.08	79.90 $\pm$ 6.95
1	Shared AE with local radius	100.00	51.47	137.83
2	Asset-specific early-life AE	100.00 $\pm$ 0.00	17.69 $\pm$ 2.56	33.03 $\pm$ 10.80
2	Shared AE with local radius	100.00	16.86	46.83
3	Asset-specific early-life AE	100.00 $\pm$ 0.00	10.63 $\pm$ 0.00	15.00 $\pm$ 0.00
3	Shared AE with local radius	100.00	10.29	15.50
4	Asset-specific early-life AE	100.00 $\pm$ 0.00	39.37 $\pm$ 1.26	63.63 $\pm$ 5.89
4	Shared AE with local radius	100.00	35.41	132.83
Mean	Asset-specific early-life AE	100.00	30.19	47.89
Mean	Shared AE with local radius	100.00	28.51	83.25

The results do not establish that the shared representation is universally superior as a prognostic detector, since alarm timing depends on the representation, persistence criterion, and the particular degradation trajectory. They do show, however, that eliminating per-asset representation learning produced no observed loss of departure detection in this experiment. A single simulator-trained representation, combined with local operational-radius calibration, matched the monitoring performance of twenty independently trained asset-specific models while producing comparable—or earlier—persistent-departure timing.

From a deployment perspective, the difference is substantial. The asset-specific reference requires fitting and maintaining an independent scaler and neural representation for every monitored unit, whereas the proposed methodology requires one centrally maintained representation and only lightweight local calibration at deployment.

5.8 Edge deployment implications

The shared model contains 847,601 trainable parameters and is trained entirely offline. Deployment at the monitored asset requires only forward inference, a small residual buffer, estimation of an empirical quantile, and the current convergence state. No gradient computation, model optimization, fault labels, or asset-specific representation learning are required after installation.

On the evaluation platform, inference over a 1,200-sample window requires a median CPU latency of only 6.26 ms (95th percentile: 6.62 ms), substantially below the 100 ms acquisition interval corresponding to the evaluated windows. Consequently, the proposed deployment operates comfortably in real time while leaving ample computational margin for additional edge tasks such as communication, diagnostics, or signal preprocessing.

The local deployment state is also extremely compact. Besides the shared model itself, each asset stores only a buffer of the most recent 100 reconstruction residuals (400 bytes in float32) and a few scalar calibration variables, requiring less than 1 kB of asset-specific state. The only fleet-wide artifact is the shared nominal representation, which is trained once and distributed to every operational unit.

Table 10: Edge-deployment footprint of the proposed shared nominal representation.

Metric	Value
Model parameters	847,601
Raw float32 weight size	3.39 MB (3.23 MiB)
Serialized inference model (.keras)	10.24 MB (9.76 MiB)
CPU inference latency (median)	6.26 ms / 1,200-sample window
CPU inference latency (95th percentile)	6.62 ms / window
Local residual-buffer length	100 samples
Residual-buffer memory	400 bytes
Total local calibration state	428 bytes (< 1 kB)
Per-asset model training	None
Fault labels required	None

6 Mathematical Scope and Deployment Guarantees

The deployment methodology presented in this paper is intentionally engineering-oriented. Its mathematical foundations are developed in full in our accompanying technical reports on inverse recovery and quotient-space formulations Di Santi 2026a; Di Santi 2026b. The purpose of this section is not to reproduce those developments, but to state the minimal functional-analytic scaffolding that justifies why a single shared canonical nominal representation and independently estimated local operational radii are mathematically compatible, and to make explicit the computational trade-off that motivates the condensed proof structure adopted here.

Mathematical Scope and Computational Complexity. Jointly recovering a nuisance-invariant canonical representation *and* calibrating per-asset decision boundaries across an unbounded, heterogeneous fleet is combinatorially hard in general, since it requires a simultaneous search over the unknown nuisance action G_{nuis} and the per-asset acceptance parameters $\{\tau_k\}_{k=1}^K$. This paper avoids solving that joint problem directly by decoupling it into two tractable sub-problems: (i) a *global* inverse problem, solved once and offline, recovering a canonical nominal representation whose orbit is relatively compact under Proposition 1; and (ii) a *local* inverse problem, solved independently and in real time by each deployed asset, consisting of a one-dimensional empirical-quantile estimation with $O(1)$ update cost per observation. The proofs of Propositions 1–2 are presented in condensed form because their role in this paper is to establish *existence and factorization guarantees* that justify this engineering separation, not to derive new functional-analytic results; complete derivations, including verification of the equicontinuity modulus for the physics-based nuisance simulator and the full quotient-space construction, are provided in the companion technical reports Di Santi 2026b; Di Santi 2026a, consistent with the space constraints of an applied transportation venue.

6.1 Observation model

Let $X \subset \mathbb{R}^n$ denote the space of fixed-length observation windows, $\theta \in \Theta$ the latent physical (diagnostic) state, and $\eta \in E$ the nuisance variables associated with sensor coupling, mounting, operating point, and environmental noise. The observation process is

$$x = F(\theta, \eta). \quad (14)$$

Let G_{nuis} denote a physically motivated family of nuisance transformations that act on E (or equivalently on X) while preserving the underlying diagnostic state θ . Two observations are

therefore considered *diagnostically equivalent* whenever they differ only through the action of G_{nuis} , i.e., $x \sim x'$ iff $x' = F(\theta, g \cdot \eta)$ for some $g \in G_{\text{nuis}}$ and the same θ .

6.2 Existence of a compact nominal representation

Proposition 1 (Relative compactness of the nominal orbit). *Let $\theta_N \in \Theta$ denote the nominal (healthy) latent state. If E is compact and the map $\eta \mapsto F(\theta_N, \eta)$ is uniformly bounded and equicontinuous on E in the sup-norm topology, then the nominal orbit*

$$O_N = \{F(\theta_N, \eta) : \eta \in E\} \subset C(X) \quad (15)$$

is relatively compact in $(C(X), \|\cdot\|_\infty)$.

Proof (condensed). Since E is compact and $\eta \mapsto F(\theta_N, \eta)$ is continuous on E , the family $\{F(\theta_N, \eta) : \eta \in E\}$ is pointwise bounded, and by hypothesis it is equicontinuous under the sensor-response regularity assumed for F . The Arzelà–Ascoli theorem Rudin 1976 then guarantees that O_N is relatively compact in $C(X)$ equipped with the uniform norm; consequently its closure $\overline{O_N}$ is compact. Full verification of the equicontinuity modulus for the physics-based nuisance simulator used in Stage 1 (Section 4) is given in the companion technical report Di Santi 2026b; it is omitted here under the scope justified in the box above. $\square$

A representation that collapses nuisance-equivalent observations therefore maps nominal operation into a compact canonical structure. The shared representation learned offline (Section 3.2) approximates this structure, while the operational radius $\tau_k^\star$ accounts only for the asset-specific numerical scale and dispersion of the anomaly score around it. Estimating $\tau_k^\star$ does not modify the representation itself; it identifies a bounded local acceptance neighborhood around the same fixed canonical nominal object. This is precisely the sense in which the empirical convergence of $\tau_k(t)$ observed after 51.0 ± 12.6 windows on CWRU (Section 3.4) is not a heuristic artifact but the expected behavior of a locally identified radius drawn from a bounded, equicontinuous family of nuisance-induced residual distributions: because O_N is relatively compact, the residual statistics available to every asset are drawn from a bounded family, which is what permits rapid, finite-sample convergence of $\tau_k(t)$ rather than an unbounded or divergent search.

6.3 Factorization of invariant diagnostic maps

Proposition 2 (Factorization through the nuisance quotient). *Let $\pi : X \rightarrow X/G_{\text{nuis}}$ denote the canonical quotient projection induced by the equivalence relation of Section 6.1. If a diagnostic map $d : X \rightarrow \mathbb{R}$ is invariant under the action of G_{nuis} , i.e., $d(g \cdot x) = d(x)$ for all $g \in G_{\text{nuis}}$ and $x \in X$, then d is constant on every equivalence class $\pi^{-1}([x])$ and there exists a unique map $\bar{d} : X/G_{\text{nuis}} \rightarrow \mathbb{R}$ such that*

$$d = \bar{d} \circ \pi. \quad (16)$$

Proof (condensed). Invariance of d under G_{nuis} implies d is constant on each fiber $\pi^{-1}([x])$ of the quotient projection. The universal property of the quotient map then yields a unique $\bar{d}$ satisfying Eq. (16); existence follows by defining $\bar{d}([x]) := d(x)$ for any representative x of the class $[x]$, which is well defined precisely because d is constant on fibers. The full characterization of G_{nuis} -orbits under sensor-coupling and operating-point variability, and the topology inherited by X/G_{nuis} , is treated at length in Di Santi 2026b; this paper uses only the factorization consequence stated above. $\square$

Proposition 2 provides the geometric justification for treating diagnosis as membership testing around a canonical nominal representation rather than performing classification directly on raw observations x . Within this interpretation, representation learning solves the global inverse problem exactly once, recovering an approximation of $\bar{d}$ (equivalently, of the compact orbit O_N

from Proposition 1) offline and centrally. Deployment, by contrast, solves only a lightweight *local* inverse calibration problem for each physical asset k : given the fixed representation, estimate the residual quantile τ_k^* that realizes a prescribed nominal coverage $1 - \alpha$ (Eq. (4)) for that asset’s own nuisance-induced residual distribution. Because Proposition 1 guarantees that the underlying orbit is bounded and Proposition 2 guarantees that the diagnostic map is well defined on the quotient, this local calibration is a well-posed, finite-dimensional estimation problem rather than a re-solving of the global representation-learning task—precisely the mathematical structure invoked as “Day-1” deployment in Section 1: the map from raw field observations to an operational decision boundary is a bounded, topologically justified calibration, not a heuristic shortcut.

This mathematical structure is directly reflected in the empirical results of Sections 5.3 and 5.7. On CWRU, the same fixed representation transfers across four distinct operating conditions (1730–1797 RPM) while only the local radius $\tau_k(t)$ is re-identified, yielding 99.69% average ball-fault detection and 100% inner- and outer-race detection under online calibration. On the NASA IMS run-to-failure dataset, the identical principle extends from controlled operating-condition heterogeneity to physically distinct bearing units, yielding persistent-departure detection in all four bearings with an average lead time of 83.25 h. Both results are consistent with the claim that the representation—not the absolute residual scale—is the transferable mathematical object, exactly as predicted by Propositions 1–2. A complete mathematical treatment of the inverse-problem formulation, the quotient-space construction, and their relation to canonical nominal representations is provided in Di Santi 2026a; Di Santi 2026b.

7 Discussion

The principal result of this work is not that a Conv1D autoencoder can separate CWRU fault classes. That capability has already been extensively demonstrated in the literature. The contribution lies instead in the deployment methodology: a single canonical nominal representation can be shared across heterogeneous assets, while each physical unit independently estimates its own operational radius without retraining the representation.

The repeated-run experiments provide an additional insight into this deployment strategy. Independent random initializations converge to slightly different numerical reconstruction-error scales, yet the resulting healthy false-alarm rate and fault detection performance remain essentially unchanged after empirical calibration. Different stochastic realizations therefore learn diagnostically equivalent nominal representations even though their absolute residual scales differ.

A complementary robustness experiment leads to the same conclusion from the deployment side. Keeping the representation fixed while varying only the nominal observations used during the 100-window warm-up produces only modest variations in the estimated operational radius and little change in post-calibration detection performance. The deployment mechanism is therefore robust at two independent levels: the learned representation itself and the local calibration procedure.

These observations reinforce the central claim of the paper. The transferable object is not an absolute reconstruction-error threshold but a shared canonical nominal representation. The role of local calibration is simply to identify the residual value that realizes the common target nominal coverage for each individual installation.

The calibration-transfer experiment further shows that the representation transfers successfully from simulation to field deployment, whereas the absolute reconstruction-error scale does not. A fixed simulator threshold therefore cannot be deployed fleet-wide. What transfers is the representation together with the target nominal coverage, while each asset estimates its own local operational radius from its residual distribution.

This deployment methodology also fits naturally within the broader theoretical framework

developed in our previous work. In our earlier technical report Di Santi 2026b, industrial diagnosis was interpreted as recovering a canonical nominal representation from observations rather than directly classifying raw signals. That interpretation was subsequently formalized as an inverse problem over quotient spaces Di Santi 2026a, where the canonical representation constitutes the mathematical object to be recovered while nuisance variability is factored out.

The present work addresses the deployment stage of that formulation. Once the shared canonical representation has been learned offline, each deployed asset no longer needs to solve the global representation-learning problem. Instead, deployment consists only of solving a lightweight inverse calibration problem: given the reconstruction residuals generated by the fixed shared representation, each asset estimates the operational radius that best characterizes its own nominal operating conditions.

This viewpoint is also consistent with the blueprint formulation introduced in Di Santi 2025, where diagnosis was interpreted as recovering a compact nominal structure from observations. Here, that recovered canonical representation is assumed to be available, and only its local realization must be identified for each deployed asset. Consequently, representation learning solves the global inverse problem only once, whereas deployment repeatedly solves a lightweight local inverse calibration problem for every physical asset.

Operational-radius convergence also clarifies the role of adaptation. The objective of adaptation is to identify the asset during cold start, not to follow its aging indefinitely. Once τ_k^* has been established, continued movement of the anomaly-score distribution constitutes diagnostic evidence. Allowing unrestricted adaptation after convergence would instead risk absorbing progressive degradation into the nominal baseline.

The observed variability in healthy false-alarm rate should be interpreted in the context of the deliberately short initialization interval. Using 1,200-sample non-overlapping windows acquired at 12 kHz, the 100-window warm-up corresponds to only approximately ten seconds of operation. Estimating an empirical 98th percentile from such a short sequence inevitably depends on a very small number of extreme observations.

A practical deployment would not necessarily terminate initialization after a fixed number of windows. Instead, the operational radius could continue to evolve automatically until the convergence criterion is satisfied, or until a minimum amount of nominal operating time has been observed. Once convergence is achieved, the radius can be frozen or tightly constrained so that subsequent degradation cannot redefine itself as nominal.

From an industrial perspective, the proposed architecture offers several practical advantages. It enables day-one deployment without fault labels, removes per-asset representation learning, isolates calibration between physical assets, requires only lightweight edge computation, and reduces continuous transmission of high-frequency sensor data.

An additional consequence is that computational scalability becomes decoupled from fleet size. Fleet growth increases the number of monitored assets, but not the number of engineered models. Adding a new operational unit requires only distributing the shared representation and estimating its local operational radius. Representation engineering, validation, and maintenance remain constant regardless of the number of deployed assets.

The computational requirements further support this deployment model. On the evaluation platform, inference requires only 6.26 ms for a 1,200-sample window, corresponding to approximately 6% of the available processing budget for the evaluated acquisition rate (100 ms per window). The system therefore provides roughly a sixteen-fold real-time processing margin while requiring less than 1 kB of asset-specific deployment state beyond the shared representation.

An additional practical advantage is that calibration is performed around a previously learned canonical nominal representation rather than around the deployment data themselves. Asset-specific retraining implicitly assumes that the calibration dataset is entirely healthy. If the monitored asset already exhibits incipient degradation, that degradation may become part of the learned nominal model. In contrast, the proposed methodology preserves the shared

canonical representation and estimates only a local operational radius. This separation opens the possibility of identifying assets whose initial behavior is already inconsistent with the canonical nominal manifold before they are accepted as locally healthy.

7.1 Future Work

The present work intentionally focuses on day-one deployment under the assumption that the monitored asset begins operation in a nominal condition. A natural extension is to relax this assumption.

Because the methodology separates the shared canonical representation from the asset-specific operational radius, future work will investigate whether the initialization phase itself can be validated against the canonical nominal manifold. Such a consistency test could distinguish genuinely healthy initialization from assets already exhibiting incipient degradation, thereby preventing abnormal behavior from becoming part of the local baseline.

A second research direction concerns progressive physical degradation. Rather than detecting only departures beyond a converged operational radius, future work will investigate whether the joint evolution of reconstruction residuals and operational-radius dynamics contains predictive information about the degradation process. This would naturally connect the proposed deployment methodology with physics-informed health indicators and remaining useful life estimation.

Finally, the mathematical interpretation introduced in our previous work Di Santi 2026a opens several theoretical questions. If the shared canonical nominal representation is indeed the transferable mathematical object, then the operational radius may be interpreted as defining an asset-specific neighborhood around that object. Formalizing this relationship, characterizing its invariance properties, and establishing conditions under which local calibration preserves the underlying quotient-space structure constitute promising directions for future research.

8 Conclusion

We presented a day-one deployment methodology for nominal condition monitoring across heterogeneous asset fleets, validated on public rolling-element bearing datasets. A single nominal representation is trained once on simulated healthy observations and transferred unchanged to field deployment. Each operational unit then performs only a short unlabeled warm-up to estimate its own operational radius, without modifying or retraining the shared representation.

The experimental results show that a simulator-trained nominal representation transfers successfully to field deployment, whereas an absolute decision threshold does not. A shared target nominal coverage, realized locally through the empirical residual quantile estimated during initialization, provides an effective bridge between the shared representation and field deployment. This deployment strategy preserves detection performance close to that of asset-specific models trained under ideal asset-specific calibration conditions while eliminating per-asset representation learning.

The transferable object is therefore not an absolute anomaly threshold, nor an asset-specific representation, but a shared canonical nominal representation together with a lightweight local calibration mechanism. The central outcome of this work is consequently not a new anomaly detector, but a scalable deployment principle: learn the canonical nominal representation once, distribute it across the fleet, and let each physical asset identify only its own operational radius during installation.

This separation between representation learning and deployment calibration provides a practical foundation for large-scale edge condition monitoring in future transportation systems.

Such a representation could naturally be produced and maintained by the original equipment manufacturer (OEM), while fleet operators perform only lightweight local calibration during deployment, thereby separating representation engineering from operational deployment.

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